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MILITARY STANDARD

TEST AND INSPECTION OF CRANE, HYDRAULIC, MOBILE, DIESEL RUBBER TIRES

FSC 3950

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DEPARTMENT OF DEFENSE
WASHINGTON, D C

TEST AND INSPECTION OF CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES MIL-STD-261C

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commanding Officer, Navy Ships Parts Control Center, Box 2020, Code 0302, Mechanicsburg, PA 17055-0788) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TEST AND INSPECTION OF CRANE,

HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES.

1. SCOPE

1.1 Scope. This standard covers the testing and inspection of cranes listed in Specification MIL-C-3998C

2. REFERENCED DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this standard.

SPECIFICATIONS
MILITARY

MIL-C-3998--- CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES.

PUBLICATIONS

MILITARY

Storage and Materials Handling Manual

Department of Army Technical Manual TS-743-200-3

Department of Navy NAVSUP Publication 284, Vol.II

Department of the Air Force AFM 67-3, Vol.II

U. S. Marine Corps Joint Publication NAVMC-1101, Vol.II

Defense Supply Agency Manual DSAM 4145.1,Vol.II

(Copies of publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS. Not applicable.

4. GENERAL REQUIREMENTS

4.1 Requirements for approval. Items of equipment tested and inspected under this standard shall pass the following tests and inspections outlined herein:

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A. Pre-Test

B. Performance Test

C. Post Test

Examination and testing, whenever practicable, shall be conducted at the contractor's plant. If the contractor does not have an approved test course, the endurance tests and post-tests may be conducted at a Government test course at contractor's expense. All pre-inspections and pre-tests shall be conducted at the contractor's plant and a copy of the results of such test shall be forwarded by the contractor with the truck to the designated government testing facility. The test facility will then complete the endurance and post-tests and forward the results to the appropriate procuring activity for final review and approval. A copy of the results will be forwarded with the truck to the cognizant Government Quality Control Representative.

4.2 Crane truck operators.

4.2.1 Operator personnel. When equipment is tested at a Government facility, equipment operators shall be furnished by the government to assist the Test Officer in conducting the test and inspections as outlined in this standard. The contractor may supply their own operator for the test unit at their own expense.

4.2.2 Operator qualification. Operators assigned for testing the equipment under the provisions of this standard shall have completed satisfactority the applicable portions of the course entitled "On-The-Job Training Course for Operators of Materials Handling Equipment - Joint Storage and Materials Handling Manual". Operators shall have available and show to contractor's representatives, if requested, the "Standard Operator's Permit" (DD form 313), or its equivalent, for the type of equipment under test.

4.2.3 Operator indoctrination. Operators provided for testing any truck under provisions for this standard shall be required to drive the truck through the appropriate test course until sufficient proficiency is achieved in operating the truck. This warmup period shall last for not more than 2 hours per driver. The contractor's representatives will be permitted to inform each driver of the proper technique to be used in operating the truck to achieve maximum performance.

4.3 Test location. The test location will be designated by the Contracting Officer. When the contractor's approved test course is designated as the test location, the contractor's preferred equipment operators may be used.

4.4 Test Officer. The test officer shall be appointed by the Department of Defense and shall exercise full authority in directing the test.

4.5 Ambient Temperature. Temperature of the surrounding atmosphere may range from 40°F. to 104°F. inclusive. If the temperature drops below or rises above this range, the test may be discontinued at the discretion of the test officer and the manufacturer until the temperature returns to this range. The Government assumes no responsibility for adverse affects that may result from testing in other than the range of temperatures specified herein.

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4.6 Contractor Participation. Any contractor whose truck is being tested in accordance with the provisions of this standard will be permitted to have representatives witness the test and inspection and assist the test officer (upon request).

4.7 Official Test Report. An official report of the tests and examinations shall be prepared by the designated testing facility. It shall include records of the pretest, endurance tests, post-tests, and such other tests as were deemed necessary together with a statement from the test officer indicating whether the machine has passed. The number of copies required shall be directed by the contracting officer.

5. DETAILED REQUIREMENTS

5.1 Pre-Test. The pre-test shall consist of 3 principal parts, each of which is described below:

A. Maintenance Service and Inspection

B. Inspection Check List

C. Inspection Tests

5.1.1 Maintenance Service and Inspection Procedures. Maintenance Service and Inspection shall be conducted in accordance with the applicable Maintenance Service and Inspection Sheet listed hrerin.

5.1.2 Inspection Check List. All inspections and other procedures needed to answer the questions contained in the check list shall be performed in conjunction with, or immediately subsequent to, the initial maintenance service and inspection test referred to in 5.1.3. Each of the questions contained in the pre-test check list shall be answered by placing an "X" in the column headed "Yes" or "No" as appropriate. After completion of the check list, any "X" in the "No" column shall be cause for rejection of the truck under test.

5.1.3 Inspection test. Inspection tests shall be conducted in accordance with the procedures outlined in the test forms to determine compliance with contractual requirements. Each of the questions contained in the test forms shall be answered by placing an "X" in the column headed "Yes" or "No" as appropriate. After completion of these additional inspections, all test forms will be examined. Any "X" in the "No" column of one or more of these forms shall be cause for rejection, subject to correction, of machine under test.

5.1.4 Procedures. All portions of the pre-test shall be completed satisfactorily at the contractor's plant prior to conducting the endurance test. When endurance testing is conducted at a Government test facility, pre-tests not conducted at the contractor's plant will be conducted by the Government upon request from the contractor through the cognizant representative at contractor's expense. Inspections and tests under the provisions of this standard requiring the use of standard loads shall utilize the appropriate load detailed in Figure 1 or a load of another composition and construction having the same weight, location of center of gravity, and safety features as the applicable load detailed on Figure 1. A tolerance of - 1% + 0% of the specified test loads will be permitted. Loads for the stability tests shall be as detailed in Figure 1,

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i.e. 10% in excess of rated load.

5.2 Endurance test.

5.2.1 Layout of course. The test course shall be set up in accordance with the provisions of Figure 2. Aisle widths for all sizes shall be eleven (11) feet. The entire perimeter of the test course shall be encompassed with suitable materials to form a barrier having a height of at least 12 inches. Rubber pylons, or other suitable marking devices, shall be placed at all corners and other critical points to assist in keeping the truck under test within the course aisle.

5.2.2 Obstacle course. The obstacle test shall be set up in accordance with Figure 3, as a part of the test course. Iron or steel blocks may be used. A guide line 15 feet long shall be provided to the left of the center line of the course and paralled to it. The inner edge of this guide line shall be at a distance from the course center line equal to 1/2 of the truck width, plus 6 inches. This spacing and arrangement will allow the left wheels of the truck under test to pass over the first pair of blocks and then the right wheels to pass over the second pair of blocks while the truck is moving forward in a straight line parallel to the guide line.

5.2.3 Ramp design. The ramp shall be set up in accordance with Figure 2 as a part of the test course.

5.2.4 Test loads. Rated loads used in the performance test shall be in accordance with paragraph 5.1.4.

5.2.5 Load areas. Loads in Positions 1, 2, and 3 shall be located in accordance with Figure 2. They shall be placed in frames made of 6 by 6-inch timbers, which have an inside measurement of 52 by 52 inches. Frames shall be attached to the floor in the positions shown. Test loads in Positions 4, 5, and 6 shall be located in accordance with Figure 2. They shall be placed in frames made of 6 by 6-inch timbers, which have an inside measurement of 52 by 52 inches. Frames shall be attached to the floor in the positions shown.

5.2.6 Lighting requirement. Lights of the truck under test shall be on at all times during the test. The headlights shall be turned off and on at the beginning of each lap and at each loading area.

5.2.7 Course surface. Except where otherwise specified, the course shall be paved with concrete or the equivalent, having a fine float finish.

5.2.8 Speed of negotiating course.

5.2.8.1 General. Each truck undergoing test should traverse each part of the performance test course at the maximum safe practical speed.

5.2.8.2 Duration of endurance test. Each truck shall complete 1200 circuits of the test course. The endurance test shall be conducted on a continuous basis during each operating 8 hour day with 7 circuits being equivalent to 1 hour of operation. Refueling shall be permitted as necessary to permit continuous operation for each operating day. However, fuel tank capacity shall be such as to allow for a minimum of 56 complete circuits without

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refueling. One half the total number of circuits shall be traversed in the opposite direction, i.e. 50% of the circuits shall be accomplished in the clockwise direction and 50% in the counterclockwise direction.

5.2.9 Periodic Maintenance. Prior to the endurance test and after completion of each 300 circuits of the endurance test, maintenance and inspections shall be performed in accordance with the applicable Maintenance Service and Inspection sheet. No repairs or replacement of parts shall be permitted at this time.

5.2.10 Method of negotiating course. The crane under test shall begin each lap at the point labeled "START" Figure 2. The operator shall sound the horn for approximately one second at start of test. In lap number 1 it shall proceed forward along the portion of the course marked "A" until it reaches the first load area. The applicable rated load shall be placed in this location. The boom of the crane shall be slued 90° to the right to remove the test load from Position 1. The boom shall then be brought in line with crane travel and the crane shall proceed along the main aisle marked "B" until it reaches Position 2. Size C cranes, in addition to complying with sluing requirements, shall deck all loads for point to point transit. All other style cranes may secure the load with guy lines or an alternate method of preventing swinging of load while in transit. The crane under test shall slue 90° to the left and set down the test load in Position 2. The boom shall then be brought in line and proceed forward along the main aisle marked "C" until it reaches Position 3. The boom shall be slued 90° to the right and pick up the test load in Position 3. The crane shall then proceed forward with the boom in line along the main aisle marked "D" until it reaches the obstacle course marked "E". The obstacle course shall be traversed in a straight line paralled to the guide line so that its left wheels pass over the first pair of blocks and its right wheels pass over the second pair of blocks. This part of the test may be conducted at low speed with the transmission in low gear. However, the crane shall not be stopped prior to or while going over the blocks. The crane shall then proceed farther along the main aisle marked "F" and "G" until it reaches the position on the ramp labeled "Stopping Point" on the drawing. Here the operator shall stop the crane, shift gears, and then resume mounting the ramp. When reaching the top level of the ramp, the crane shall be shifted to high gear and proceed along the main aisle marked "H", "I", and "J". Load may be held at load center by guy lines or an alternate method to prevent load from swinging out beyond load center, when traversing down the ramp. When the crane reaches the portion of the course marked "K" the operator shall stop it and then restart it. After completion of this portion of the test he shall proceed along the main aisle to the point marked "Stop". All stops shall be at the maximum safe deceleration rate. In the second lap the crane shall re-enter the portion of the course marked "A" on the drawing. In portions "A", "B", "C" and "D" the procedures shall be the same as described for lap number 1, except that the crane shall set down a test load at each point where previously it was indicated to pick one up and pick up a test load where previously it was indicated that one should be set down. Maneuvers in the balance of the course, and the order in which the course is covered beyond the portion marked "D" shall be the same as described for lap number 1. Lap number 3 shall be identical with lap number 1, except that Position 4 shall be substituted for Position 1, Position 5 for Position 2, and Position 6 for Position 3. Also the indicated direction of sluing at each of the 3 positions inlap number 1 shall be reversed in lap number 3. Lap number 4 shall be identical with lap number 2, except that references indicating the difference of

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lap number 2 to lap number 1 should refer to lap 4 from 3. The operator shall continue to maneuver the crane around the course, following the pattern described above for lap numbers 1, 2, 3, and 4 in accordance with paragraph 5.2.8. The operator of the crane shall keep the boom in the direction the crane is traveling (in line) and within the permissible aisle limits at all times while traversing the test course. The "Hourly Time Record Sheet" shall be filled out for each operating hour of the performance test.

5.2.11 Failure during endurance testing. During the endurance testing, minor adjustments or repairs will be permitted at the discretion of the test officer. Minor breakdowns which are repetitive (to be determined by inspection service) and major breakdowns shall be cause for rejection of the crane. A major breakdown is defined as a failure in a major component of the crane or one requiring more than one hour to repair using conventional shop tools and practices.

5.3 Post-Test. The post-test shall be conducted after completion of the endurance test; or if the crane fails the endurance test, the inspection portion of this procedure shall be conducted after such failure. The post-test consist of 2 parts.

A. Post-Test inspection tests

B. Post-Test inspection.

5.3.1 Post-Test inspection tests. These tests shall be performed in accordance with the applicable procedures provided in the test forms. The crane should be given the maintenance service and inspection in accordance with the applicable Maintenance Service Inspection sheet. Each of the questions contained in the Post-Test Inspection Test Check Sheet shall be answered by placing an "X" in the column headed "Yes" or "No," as appropriate. After completion of the Post-Test Inspection Test Check Sheet, any "X" in the "No" column will be cause for rejection of the crane under test.

5.3.2 Post-test inspection form. Personnel under the direction of the test officer shall examine the crane and make applicable notation in the post-test inspection form.

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PRE-TEST INSPECTION CHECK LIST

CRANE, HYDRAULIC, MOBILE, DIESEL,
RUBBER TIRES

Description of Crane

Crane Mfr. _____	Date _____
Model _____ Serial No. _____	Test Location _____
Recorded by _____	

1. Are all certificates of compliance furnished? YES _____ NO _____

2. Does the crane comply with the applicable safety requirements specified in ANSI-B30.15? YES _____ NO _____

NOTE: Where there is conflict between the Underwriters' Laboratories safety requirements and the specification requirements referenced in the contract, the specification requirements shall govern. Presence of the UL label attached to the crane will be considered as self-evidence of compliance with the UL standards.

3. Do the following components of the crane comply with all requirements of the governing specification?

	YES	NO
A. Engine and Engine Accessories		
Alternator	_____	_____
Regulator	_____	_____
Starting System	_____	_____
Fuel System	_____	_____
Tank	_____	_____
Lines	_____	_____
Pump	_____	_____
Air Cleaner	_____	_____
Cooling System	_____	_____
Fan	_____	_____
Water Pump	_____	_____
Thermostat	_____	_____
Radiator	_____	_____
Governor	_____	_____
Exhaust System	_____	_____
B. Transmission and Drive Assembly	_____	_____
Drive Gears	_____	_____
Axles	_____	_____
Shafts	_____	_____
Clutch (when used)	_____	_____
Fluid Coupling (when provided)	_____	_____
Torque Converter (when required)	_____	_____
Differential	_____	_____
Lubricating Fittings and Plugs	_____	_____
Transmission Type, Mechanical,	_____	_____
Power Operated	_____	_____

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C. Hydraulic System
- Pump
- Sump (Storage) Tank
- Powered Boom Extension
- Lines
- Breather and Filler Cap
- Piston Rods
- Strainer
- Filter

D. Chassis and Frame
- Bumpers
- Boom Support
- Boom Pivot

E. Load Line Hoist Mechanism
- Boom
- Cable Storage Drum
- Sheaves
- Cable
- Cable Clips
- Block

Boom Hoist Mechanism
Boom Sluing
Limit Stops
Wheels
Tires
Tubes
Valve Stem Location
Steering Wheel Mounting
Rotation of Hand Wheel
Adjustment
Power Steering
Service Brakes
Wheel Cylinders
Brake Lining
Adjustment Accessibility
Master Cylinder and Pedal
Linkages
Mechanical Parking Brake
Hand Lever
Linkages
Adjustment
Seat
Cowl (Hood)
Instrument Panel
Fenders and Running Boards
Towing Hooks
Hand Holds
Closed Cab
Cab Heater
Lifting Attachments
Controls
- Weather Protection

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Position Markings	____	____
Headlight Switch	____	____
Spotlight Switch	____	____
Ignition Switch	____	____
Starting Motor Switch	____	____
Brake Pedal	____	____
Acceleration Control	____	____
Directional Speed Controls	____	____
Boom Extension Controls	____	____
Boom Topping Controls	____	____
Boom Slewing Controls	____	____
Load Line Control(s)	____	____
F. Instruments (weather protection)	____	____
Ammeter	____	____
Oil Pressure Gage	____	____
Temperature Gage	____	____
Hour Meter	____	____
Fuel Gage	____	____
Over Temperature Signal (when required).	____	____
Headlights	____	____
Spotlights	____	____
Stoplight	____	____
Cables and wiring	____	____
Battery	____	____
Battery Mounting	____	____
Battery Cable Terminal	____	____
Horn	____	____
Fire Extinguisher (when specified)	____	____
Turning Radius	____	____
Hook Clearance	____	____
Under Clearance	____	____
Radio Interference Reduction	____	____
Nameplate	____	____
Load Capacity Chart	____	____
Instruction Plate	____	____
Registration Data	____	____
Maintenance, Service, Repair and Parts Manual	____	____
G. Other Requirements	____	____
Drain Plugs	____	____
Grease Fittings	____	____
Treatment and Painting	____	____
Access Openings	____	____
Finish and Appearance	____	____
Rivet Connections	____	____
Welds	____	____
Castings	____	____

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MAINTENANCE SERVICE AND INSPECTION SHEET

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

The following inspections and checks shall be made prior to endurance testing and prior to the post testing. Items 1 through 13 shall be checked after completion of each 800 circuits of the endurance test.

1. Check all instruments for proper operation, check free play in steering and check horn operation.

2. Test brake pedal travel, fill master cylinder, adjust brakes if necessary, report if fluid is below 1/2 full, inflate tires to proper pressure (cold) and remove all foreign matter from tires.

3. Tighten clutch pedal play, adjust if necessary.

4. Inspect all lights, windshield wipers and mirror. (When applicable.)

5. Fill radiator, check and tighten radiator hose connections, check anti-freeze in season.

6. Inspect fan and hydraulic pump belts, adjust if necessary, use ruler or straight edge (1/2" play).

7. Check engine for water and oil leaks, check filler caps, oil lines, gasket and muffler.

8. Visually check battery, check water level, check for dirty or acid condition, record readings.

9. Check fuel filter and fuel pump screens and bowls, clean as necessary.

10. Check engine oil dipstick, add oil if necessary -- dissemble air-cleaner and wash, refill with oil -- breather cap, remove and wash.

11. Check hydraulic oil dipstick, add oil if necessary. Check 25 micron filter.

12. Inspect all safety equipment -- fire extinguishers -- limit stops -- safety hook for wear or failure.

13. Lubricate chassis, using manufacturer's reference charts; replace broken fittings, change oil filter if necessary, lubricate: generator, starter, carburetor linkage, distributor shaft.

14. Drain engine oil, flush crankcase, refill with oil. Remove oil filter cartridge -- clean filter -- replace with fresh cartridge.

15. Inspect boom, boom extension, winch and load line drum for wear and damage.

16. Inspect load line, load line clips and load line guides for wear and damage.

17. Check tie rod, drag link, and wheel alignment of steering assembly.

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18. Check wheel bearings -- repack and adjust.

19. Inspect and tighten hub flange bolts.

20. Check springs for broken spring leaves and loose "U" bolts.

21. Check gear cases and fluid drive and fill all units -- report excessive consumption.

22. Check brake linings and drums; check wheel and master cylinders.

23. Inspect all wires and terminals -- tighten and replace where necessary.

24. Check hydraulic system for noise, leaks and operation -- inspect and adjust.

25. Drain and flush radiator -- check all connections and hoses -- refill with water and anti-freeze (if"in season").

26. Inspect entire exhaust systems for leaks or breaks.

Description of Crane

Crane Mfr._____		Date _____
Model_____	Serial No._____	Test Location_____
Type _____	Total hours_____	Recorded by_____

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TEST NO. 1

CLUTCH PEDAL

CRANE, HYDRAULIC, MOBILE, DIESEL,

RUBBER TIRES

Purpose:

To determine whether the force required to depress the clutch pedal meets specifications.

Test Description:

Using a pressure type indicating scale capable of of being read to 10 pound increments, measure the pressure required to depress the clutch pedal.

Test Procedure:

1. Depress clutch pedal several times to insure proper lubrication.

2. Attach gage to clutch pedal to measure force required to depress pedal.

3. Apply pressure slowly and evenly until clutch pedal is depressed to 1 inch from the floor of the vehicle. Observe maximum gage reading and record on data form.

4. Complete data form.

DATA FORM: No. 1

Description of Crane _____	Date _____
Crane Mfr. _____	Model _____
Test Location _____	Serial No. _____
Serial No. _____	Recorded by _____
Maximum Gage Reading _____	
Question ?	
Is the force required to depress the clutch pedal 40 pounds or less?	Yes _____ No _____

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TEST NO. 2

STEERING WHEEL

CRANE, HYDRAULIC, MOBILE, DIESEL

RUBBER TIRES

Purpose:

To determine the force required to turn the hand steering wheel and ability of steering gear to withstand a specified force with steerable wheels blocked. To determine number of complete turns of steering wheel to go from extreme left to extreme right and vica versa.

Test Description:

Place the unloaded truck on a level brushed concrete surface. Using a tension dynamometer or equivalent, measure the force required to move the steering wheel.

Test Procedure:

1. Place truck on dry, level, brushed concrete with steer wheels in straight ahead position.

2. Attach an adapter pulley having a root diameter equal to the steering wheel diameter to the spokes of the hand steering wheel.

3. Apply a steady force to a dynamometer attached to the adapter pulley in a counterclockwise direction and in the wheel plane, until the wheel turns.

4. Record the maximum gage reading within the first half of the inner steer wheel angle.

5. Record the maximum gage reading at extreme ends of the steer.

6. Perform 3, 4 and 5 in a clockwise direction. After completing test requirements, Steps 1 thru 6, conduct the following:

7. Raise rear of truck and block in raised position.

8. Lock one rear steerable wheel in a straight ahead position.

9. On conventional manual or power booster steering apply through the tension dynamometer, in clockwise direction, the specified force. When all hydraulic steering is used, the relief valve shall be made inoperative and with power off, apply the force. In addition, with all hydraulic steering, the above test shall be run with the relief valve inoperative and power on. In this case the steering wheel will be turned until hydraulic slippage occurs.

10. Repeat in counterclockwise direction.

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11. Perform operations 9 and 10 four times.

12. Release the locked rear steerable wheel and lock the other rear steerable wheel in the straight ahead position.

13. Repeat operations 9, 10 and 11. After completing test requirements, steps 7 thru 13, conduct the following:

14. Turn steering wheel to extreme left until it stops. Place a chalk mark on the periphery of the steering and an other mark on an adjacent surface for reference. (Other acceptable reference methods may be used)

15. Turn steering wheel to extreme right until it stops. Count number of complete turns.

16. Repeat steps 14 and 15 going from extreme right to left.

17. Complete data form.

DATA FORM: Test No. 2

Description of crane _____ Date _____
Crane Mfr _____ Model _____
Model _____ Serial No. _____ Test Location _____
Recorded by _____

	Counter-Clockwise Direction	Clockwise Direction
Maximum gage reading first half of inner steer angle	_____	_____
Maximum gage reading at extreme ends of steer	_____	_____

Indicate whether desired results were obtained Yes _____ No _____
Force required to turn the hand steering wheel. _____
Was steering mechanism and hydraulic linkage able to withstand applied force without evidence of failure or permanent deformation? Yes _____ No _____
No. of turns from Left to Right _____ Right to left _____

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TEST NO. 3a

SERVICE BRAKE PRESSURE SENSITIVITY
TOWING HOOK
CRANE, HYDRAULIC, MOBILE, DIESEL
RUBBER TIRES

Purpose:

To determine (1), relationship between variable brake pedal pressure and drawbar pull expressed in percent of the gross weight of the crane (with rated load) while crane is being towed and (2), whether towing hook has sufficient strength.

Test Description:

Tow fully loaded crane over level dry, brushed concrete, or equivalent with engine disengaged and parking brake released. Mount a scale reading device (pressure type) or the equivalent, on the brake pedal, and attach a tension dynamometer to towing hook of crane. If the surface friction between the concrete and the crane tires is less than 0.50, the surface may be brushed with abrasion compound or other material to provide suitable coefficient of friction to prevent wheels from sliding while applying the required drawbar pull.

Test Procedure:

1. Determine gross weight of crane with rated load.

2. Attach pressure gage to brake pedal in a manner which will enable pressure to be applied to the face of the gage.

3. Attach a tension dynamometer to the front towing hook of the crane with rated load.

4. Attach one end of a block and tackle, or similar mechanical device, to the tension dynamometer, and the other end to an anchoring device.

5. With transmission in neutral, move vehicle in accordance with test description above.

6. Apply sufficient brake pedal pressure to produce a force recorded on dynamometer approximately equal to a specified percent of the total weight of crane plus rated load and record brake pedal and dynamometer readings.

7. Repeat operations 2 through 6 with dynamometer attached to rear towing eye.

8. Complete data form

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Data Form: No. 3a

Description of Crane _____		Weight of Crane _____		
Crane Mfr _____		Weight of Load _____		
Model _____ Serial No. _____		Date _____		
Test Location _____	Recorded by _____			
Brake Pedal Pressure _____	Dynamometer Reading _____			
Specified percent of total Crane weight with load _____				
Indicate whether results were within specified limits	Forward		Reverse	
	Yes	No	Yes	No
Was drawbar pull developed compatible with specified brake pedal pressure?	___	___	___	___
Was towing hook able to withstand applied force without evidence of permanent distortion or failure?	N/A	N/A	___	___

TEST NO. 3b

SERVICE BRAKE PRESSURE
CRANE, HYDRAULIC, MOBILE, DIESEL
RUBBER TIRES

Purpose:

To determine whether brake pedal and brake system have the strength required by the specifications.

Test Description: Using a scale reading device (pressure type) or equivalent, apply preassure to the brake pedal.

Test Procedure:

1. Attach the pressure gage to the brake pedal in a manner which enables pressure to be applied to the face of the gage.

2. Apply specified pressure to the gage, hold for one minute and release. Repeat this operation 10 times.

3. Complete data form.

Data Form: No. 3b

Description of Crane _____	Date _____	
Crane Mfr. _____	Model _____	
Test Location _____	Serial No. _____	
Indicate whether desired results were obtained	Yes	No
Was brake system able to withstand applied pressure without failure?	___	___

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TEST NO. 4

LOAD LINE HOOK, BOOM TOPPING, AND SLUING SPEED

CRANE, HYDRAULIC, MOBILE, DIESEL RUBBER TIRES

Purpose:

To determine whether the Crane, Load Line Hook Speed, Boom Topping Speed, and Sluing Speed complies with the applicable requirements.

Test Procedure:

1. Place a reference mark on the load line approximately two (2) feet above the load contact point of the hook. Place a second reference mark ten (10) feet above the initial mark.

2. Position crane with load off the ground with second reference mark at a conviently oriented point in the vicinity of the outer boom pulley.

3. At a given signal the crane operator lifts the load and an observer starts a timing device.

4. When the lower reference mark on the cable reaches the orientation point the timing device is stopped. Record the time in seconds.

5. Repeat the above, going from the raised position back to the ground. Perform this test three (3) times.

6. With rated load suspended and boom horizontal or at rated elevation for size A cranes, on a given signal,, operator tops the boom and observer starts timing device. When boom reaches required elevation, stop timing device. Record time in seconds.

7. Repeat Step 6 going from topped position to depressed position. Perform Step 6 without load.

8. Repeat Step 7 without load.

9. Suspend rated load from hook, placed at 90° from longitudinal center line of crane for size A and at any horizontal position for sizes B and C. On a given signal observer starts timing device and operator slues boom through its applicable degrees of slue. Stop timing device, record time in seconds, at this point. Repeat test in opposite direction. Perform entire sluing test three (3) times.

10. Complete data form.

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Data Form: No. 4

Description of Crane _____ Date _____
Crane Mfr. _____ Model _____
Test Location _____ Serial No. _____ Recorded by _____

A (Average time) = T1 + T2 + T3 (total time for 3 trials)/3 _____
B (Load line hook spped feet per minute) = 60 x 10/A _____

	T1	T2	T3	A
Time required to raise load line hook	___	___	___	___
Time required to lower load line hook	___	___	___	___
Time required to top boom	___	___	___	___
Time required to depress boom	___	___	___	___
Time required to slue boom clockwise	___	___	___	___
Time required to slue boom counterclockwise	___	___	___	___

	Yes	No
Does load line hook speed comply with specified requirements?	___	___
Does topping speed comply with specified requirements?	___	___
Does sluing speed comply with specified requirements?	___	___
Does crane slue efficiently and continuously clockwise and counterclockwise for size B and C cranes?	___	___

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TEST NO. 5

SPEED

CRANE, HYDRAULIC, MOBILE, DIESEL,RUBBER TIRES.

Purpose:

To determine speed of crane, without load, on a level surface over a measured distance.

Test Description:

Crane, without load, is accelerated to maximum speed and timed through 88 foot distance. (This 88 foot length is used to simplify calculations as 88 feet/minute is 1 m.p.h.). Shorter distances may be used when calibrated electronic devices are employed. Three runs are made forward in high gear and three in reverse. The data is recorded and average speeds determined. Hook may be secured to prevent lashing.

Course Layout:

A.....B.....88'.....C.....

Test Procedure:

1. Warm up crane for 15 minutes prior to test.

2. Locate crane, without load, on level surface with front wheels at point A.Start forward test run and have crane in high gear and at top speed by the time it reaches point B. Record time (seconds) required to travel between point B and point C. Repeat 2 times.

3. Perform operation 2 in the rearward direction.

4. Complete data form.

Data Form: No. 5

Description of crane		
Crane Mfr.	Date.	
Model.	Test Location.	
Serial No.	Recorded by .	
	Forward	Rearward
Time in seconds, No. 1 test		
Time in seconds, No. 2 test		
Time in seconds, No. 3 test		
Total time		
Average time (1/3 of total time)		
Average speed in m.p.h 60/Average time		
Indicate whether desired results were obtained. Yes	No	
Average forward speed		
Average rearward speed		

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TEST NO. 6

(a) ACCELERATION - (b) ACCELERATION RESPONSE

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose: To determine acceleration of crane, with and without load, on a level surface from a standing start; and check acceleration response.

Test Description: Crane is accelerated at maximum rate from a standing start and time through a distance as required by specification using a calibrated timing device. The acceleration response test shall be conducted as a second part of this test form. (See note.)

Course Layout:

A.....Variable .....B

Test Procedure: (Acceleration)

1. Warm up crane for 15 minutes prior to test.

2. Locate crane with the specified rated test load, with front wheels at point A. (Load shall be tied down for size A and B cranes and carried on the deck for size C cranes.

3. The crane shall be accelerated at maximum speed and time through distance A to B and recorded.

4. Repeat operation 3 a second and third time and average.

5. Repeat steps 1 thru 4 without load.

6. Complete data form.

Data Form: No. 6a

Description of Crane			
Crane Mfr.	Date		
Model	Test Location.		
Serial No			
	DISTANCE		
	A	B	
Time in seconds for No. 1 test			
Time in seconds for No. 2 test			
Time in seconds for No. 3 test			
Total time			
Average time (1/3 of total time)			
Indicate whether crane meets prescribed requirements		Yes	No
Average acceleration			
Acceleration response			

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7. Test Procedure: (b) (Acceleration Response)

8. Place crane with specified rate load on level surface and with engine running at idle speed, put transmission in low speed, forward drive. If crane has manual shift transmission, the clutch shall be disengaged.

9. At a pre-arranged signal, the operator shall depress the accelerator and the timing device shall be started simultaneously. The crane shall be observed closely. At the instant of crane movement, the timing device shall be stopped and lapsed time shall be recorded.

10. Perform this test in both forward and reverse directions.

NOTE: For this test, cranes incorporating fluid drive or other clutchless drive shall be operated for a sufficient period of time to assure crane will not creep.

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TEST NO. 7

SLOPE ASCENSION - PARKING BRAKE - UNDERCLEARANCE
CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose:

To determine whether crane can accelerate from a dead stop on a specified percent grade with and without rated load; and to determine ability of parking brake to hold crane with rated load on this grade with load on uphill side. Also, to determine under clearance.

Test Description:

Crane ascends a specified percent grade in a forward direction at low speed on a dry concrete surface. Crane stops and starts up slope again. Horizontal approaches should be provided for the ramp. Under clearance is observed during this test.

Test Procedure:

1. Drive crane with rated load along horizontal approach and onto ramp of specified percent grade and bring to complete stop. (All wheels supported on the platform.)

2. Apply parking brakes, release service brakes and observe whether crane remains stationary.

3. Release parking brake, and start from dead start and proceed up ramp and onto a horizontal surface. Measure under clearance.

4. Descend ramp. Stop on ramp and apply parking brakes, release service brakes, and observe whether crane remains stationary.

5. Release parking brake, complete descent, and proceed onto the horizontal surface.

6. Repeat without load, except for actuaing the parking brakes. Crane may be driven up ramp in rearward direction without load.

7. In lieu of driving onto a horizontal surface at top of incline, a template may be used to check underclearance.

8. Complete data form.

Data Form: No. 7

Description of Crane._____			
Crane Mfr. _____.Date._____			
Model ._____.Test Location ._____			
Serial No. _____.Recorded by ._____			
Indicate whether desired results were obtained.		Yes	No
Slope Ascension:	Loaded	_____	_____
	Unloaded	_____	_____
Underclearance		_____	_____
Parking Brake		_____	_____

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TEST NO. 8

STOPPING DISTANCE

Purpose:

To determine whether crane can be stopped within a given distance from specified maximum speeds.

Test Description:

Run crane at maximum applicable speed over dry, brushed concrete. Using a scale indicating device (pressure type), or the equivalent, apply specified brake pedal pressure and measure the distance required to stop the truck.

Course Layout: A.....Variable distance.....B.....C

Test Procedure:

1. Warm up crane for 15 minutes prior to test operating brakes at least 10 minutes.

2. Attach pressure gage to brake pedal in such a way to permit the operator to apply foot pressure to one face of the gage.

3. Run the crane unloaded in forward direction, as specified, on a strip of dry, level, brushed concrete.

4. After reaching top speed, the brakes shall be applied with the specified brake pedal pressure without disengaging the engine.

5. Measure the distance traveled by crane after application of the specified brake pedal pressure to the final stopping point. Record this distance and the brake pedal pressure.

6. Perform operations 3 through 5 a second and third time.

7. Complete data form.

Data Form No. 8

Description of Crane ._____

Crane Mfr._____.Date ._____

Model ._____.Recorded by ._____

	Brake Pressure (Pounds)	Stopping Distance (feet)
Test No. 1	_____	_____
Test No. 2	_____	_____
Test No. 3	_____	_____
Total Readings	_____	
Average readings (1/3 of total readings)	_____	
Indicate whether crane meets requirements	Yes_____	NO_____
Average brake pedal pressure	_____	
Average stopping distance	_____	

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TEST NO. 9

STABILITY

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose:

To determine whether crane has sufficient stability.

Test Description:

The applicable load for all specified load positions shall be 110% of maximum rated load. This load shall be handled in accordance with the test procedures listed below for the crane being tested. There shall be no evidence of tipping* throughout test.

Test Procedure:

1. Place test load on ground at rated load position.
2. Lift load with load line until load clears crane structure where sluing of load may make contact.
3. Slue load 90° to the right or left.
4. Hold load for one minute.
5. Slue load at 15° intervals and hold for one minute through entire slue range.
6. Complete data form.

Data Form: No. 9

Description of Crane.....		
Crane Mfr. .....Date		
Model .....Test Location		
Serial No. Recorded by		
Question	Yes	No
Were all phases of the stability test completed without tipping or loss of load?	_____	_____

* NOTE:Definition: Tipping is defined as that condition where any tire leaves contact with the supporting surface at any orientation of load as described above.

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TEST NO. 10

LIFTING ATTACHMENTS

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose:

To determine whether lifting eyes comply with specified requirements.

Test Procedure:

1. Determine the gross weight of the vehicle with reservoirs filled and record.

2. The crane shall be restrained on a level surface. Apply to each lifting eye, in a vertical direction, a static force equal to the gross weight of the vehicle as determined by step 1. The test force should be applied through a tension dynamometer with cable attachment to the device.

3. Release the test force and examine each lifting eye. The lifting eyes and crane structure shall show no evidence of deformation, fractures or permanent distortions.

4. With a four part sling lift the vehicle off the ground and hold for one minute.

5. Complete data form.

Data Form: No. 10

Description of Crane _____			
Crane Mfr. _____	Date _____		
Model _____	Test Location _____		
Serial No. _____	Recorded by _____		
Question		Yes	No
Using a 4-part sling attached to the lifting eyes, can the crane be lifted in its normal position?		____	____
Was test completed without permanent distortion of any of the lifting eyes which could be observed visually?		____	____

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TEST NO. 11

WEAR OF STEERING MECHANISM

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose:

To determine cumulative wear in the steering mechanism of the crane caused by the endurance test.

Test Description:

With the crane stationary and with no load on the hook, the amount of free play in the steering wheel is measured. This procedure is followed before and after the endurance test. If an abnormal increase in free play is noted after the endurance test is run, it is an indication of excessive wear in the steering mechanism. For cranes equipped with power steering, engine should be at idle speed.

Test Procedure:

1. Place unloaded crane on a dry, level, concrete surface with its steer wheels parallel to the longitudinal axis of the crane.

2. Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board or similar material which is semi-circular in shape and which has a radius at least 1 inch greater than the radius of the steering wheel.

3. Mark a reference point on the upper portion of the outer edge of the steering wheel; mark a corresponding point on the fiber board templet.

4. Turn the steering wheel counterclockwise until the steer wheels of the crane begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

5. Turn the steering wheel clockwise until the steer wheels of the crane begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

6. Remove the fiber board templet from the crane.

7. Place the fiber board templet on a flat surface. Measure the distance between the 2 points made in accordance with operations 4 and 5 above. This distance, in inches, is the free play of the steering wheel.

8. Check tie rod ends and other components of the steering linkage for excessive wear or instability.

9. Complete data form.

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Data Form: No. 11

Description of Crane _____		
Crane Mfr. _____ Date _____		
Model _____ Test Location _____		
Serial No. _____ Recorded by _____		
Free play of steering wheel:		
Before performance test _____ inches		
After performance test _____ inches		
Question	Yes	No
Is the free play of the steering wheel before the performance test less than 5 inches?		
Is the free play of the steering wheel after the performance test less than 1.5 times the free play of the steering wheel before the performance test?	_____	_____

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TEST NO. 12

STRUCTUAL LOAD AND OVERLOADS

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose:

To determine whether crane can withstand specified structual loads and overloads

Test Description:

Place crane on horizontal surface and support it with blocks under the axles on frame to relieve strain on tires. Determine maximum pressure required to manipulate rated loads through boom slue, topping and extension from one extreme to the other. Excessively counterweight or tie down crane to prevent tipping. Make relief valves inoperative when performing overload tests. Test in accordance with procedure outlined below.

Test Procedure:

1. With another crane, lift truck or other means, place 200% of rated load, on the hook of test crane. Hook should be at sufficient height off the ground to hold load suspended and clear truck structure when slued. Gradually provide slack on the load line of the supporting crane (or lower forks in the case of a lift truck) until test crane is supporting the entire load. Hold for 5 minutes. Slue load slowly 90°. Hold for 5 minutes. Slue load through 180°, hold for 5 minutes. Return to forward position and remove load.

2. Place rated load in a position so that it is resting against the bumper plate. With an external means move test load so that it is 3 feet from the bumper. Abrutly release allowing load to fall against the bumper.

3. Determine maximum pressure required to manipulate a rated load at specified distances thru sluing range of size A and B cranes. Apply 40% of the maximum pressure to the sluing mechanism limit stops.

4. With the crane hook positioned such that test load is suspended above the ground, using another crane or other means, place a load equal to 3 times rated load on the hook. Slowly provide slack in the load line of the supporting crane until the full load is supported by the test crane. Hold for one minute. Boom may be supported for this test.

5. With pressure relief valves set for normal operation apply percentage of maximum pressure required to manipulate the load for each function.

6. Place specified load on the deck of a Size C crane. Drive crane 400 feet.

7. Complete date form.

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Data Form: No. 12

Description of Crane .		
Crane Mfr. .	Date.	
Model .	Test Location	
Serial No.	Recorded by .	
Maximum pressure to manipulate rated load for slue		
Maximum pressure to manipulate rated load for topping		
Maximum pressure to manipulate rated load for extension		
	Yes	No
1. Does crane show any signs of damage or permanent deformation?		
2. Does hydraulic system show any signs of damage or leakage due to tests?		
3. Does bumper show any signs of damage or permanent deformation?		
4. Do load limit stops for size A or B cranes show any damage or permanent deformation?		
5. Is Size C crane capable of handling specified load without damage or permanent deformation?		
6. Do pressure relief valves open at specified pressure?		

If there are any failures, record what they are and under what conditions they occured.

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TEST NO. 13

TURNING RADIUS, AND TRAVEL CAPACITY

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose:

To determine whether crane meets specified turning radius. To determine if crane is capable of traveling short distances while carrying rated load.

Test Description:

With crane on a dry level surface drive it through 360° in a clockwise direction and 360° in a counterclockwise direction. With rated load freely suspended from the hook and crane in low speed drive the crane 100 feet.

Test Procedure:

1. Place crane on dry level surface. Attach a device to the crane at its outermost position of swing, capable of scribing a line on the level surface. Cramp steer wheels as far as possible to make crane turn in a clockwise direction. Drive crane through 360°. Measure diameter of circle scribed on ground. Repeat in counterclockwise direction.

2. Freely suspend rated load from hook. Drive crane a distance of 100 feet in slow speed.

3. Complete data form.

Data Form: No. 13

Description of Crane _____		
Crane Mfr. _____ Date _____		
Model _____ Test Location _____		
Serial No. _____ .Recorded by _____		
Diameter clockwise rotation . _____		
Diameter counterclockwise rotation _____		
	Yes	No
Is turning radius as specified?	_____	_____
Is vehicle travel with load satisfactory?	_____	_____

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TEST NO. 14

POWER TRAIN STATIC TORQUE TEST

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRE

Purpose:

To determine whether the driveline of the crane is designed to withstand specified percent of the maximum bare engine torque or stalled motor torque.

Test Description:

With drive wheels locked, the required force is applied to a torque arm attached to a tension dynamometer. The torque arm shall be attached to the input shaft of the transmission or in case of power operated drives, to the output shaft of the torque converter. This test may be performed as a bench test prior to assembly or with complete assembled crane at option of contractor.

Test Procedure:

1. Transmission shall be placed in lowest gear range.

2. Lock both drive wheels to prevent turning.

3. If transmission is the hydraulic power operated type, increase hydraulic pressure to lock clutch by external means to prevent slippage.

4. Clamp a torque arm of (L) feet to the input shaft of the transmission or in case of power operated drives to the output shaft of torque converter.

5. Obtain from manufacturer's published data, the bare engine torque (T) foot pounds of the engine. If drive includes hydraulic torque converter obtain also the stalled torque converter ratio (R).

6. Apply to the torque arm, a force equal to 1.5T/L for mechanical transmission, apply 1.5TR/L for cranes equipped with hydraulic converter.

7. Disassemble transmission and examine for failures, fractures, or permanent distortion and deformation.

8. Complete data form.

Data Form: No. 14

Description of Crane ._____		
Crane Mfr. _____ Date_____		
Model ._____ Test Location _____		
Serial No. _____ Recorded by _____		
Was crane able to withstand applied torques without evidence of failures, fractures, permanent distortion or deformation?	Yes ____	No ____
NOTE: This test is to be performed when specified either in the invitation for bid or contract.		

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TEST NO. 15

CONTROLS

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRE

Purpose:

To determine whether controls fulfill specification requirements.

Test Description:

With crane positioned on level surface operate controls as required below.

Test Procedure:

1. Place direction control lever in all drive positions (power shifted transmission only). Start crane.
2. Place direction control lever in neutral position. Start crane.
3. Actuate boom topping control for raise and depress.
4. Actuate boom extension control for extend and retract.
5. Actuate boom slue control for right then left slue.
6. Actuate load line control for hoist and lower.
7. Actuate load line control, boom topping and boom sluing simultaneously.
8. Complete data form.

Data Form: No. 15

Description of Crane ._____		
Crane Mfr._____.Date _____		
Model ._____.Test Location ._____		
Serial No._____.Recorded by _____		
	Yes	No
Does crane start in neutral only?	____	____
Does motion of controls and position of crane meet specified conditions?	____	____
Do actions take place simultaneously?	____	____
If crane fails to meet any conditions as specified, record failure._____		

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TEST NO. 16

HOOK AND CABLE

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Purpose:

To determine whether hook clearances, cable reach and paying out of hook comply with specification.

Test Description:

Place crane on level surface (preferably 5 feet above ground). Perform tests as described below.

Test Procedure:

1. Fully extend and top boom with hook in full raised position.

2. Lower hook to level surface on which crane is positioned.

3. Continue to let out cable until hook has moved another 5 feet.

4. Place crane boom and hook at positions specified. Measure distance from ground to pull point of hook.

5. Complete data form.

Data Form: No. 16

Description of Crane ._____		
Crane Mfr._____ Date _____		
Model _____ Test Location ._____		
Serial No. _____ Recorded by ._____		
	YES	NO
Indicate whether crane met specified conditions. Record any failures.	____	____

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HOURLY TIME RECORD

TRUCK MANUFACTURER	MODEL	SERIAL NO.	DATE

TEST LOCATION	RECORDED BY:	TIMER READINGS (Start)	TIMER READINGS (Finish)	REQUIRED LAPS/HR.

LAP NUMBER	TIMER READING	ELAPSED TIME	REMARKS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
TOTAL			AVERAGE LAP TIME_____

INSTRUCTIONS FOR TIMING PERFORMANCE TEST:

1. Start timing device at zero at the beginning of each operating hour.
2. Record reading at end of each lap in "Timer Reading" column.
3. Subtract reading at the end of preceding lap from reading at the end of lap. Record differences in "Elapsed Time" column.
4. If equipment under test completes the required number of laps prior to completion of an operating hour, it shall be stopped until start of next hour, and then resume performance test.

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POST TEST INSPECTION CHECK SHEET

Description of Crane _____

Crane Mfr. _____ Date _____

Model _____ Test Location _____

Serial No. _____ Recorded by _____

	Applicable test Form	Question	YES	NO
Clutch pedal	1	Does clutch pedal pressure comply with specified requirements?	____	___
Hand steering wheel	2	Does force to turn hand steering wheel comply with the specified requirements?	____	___
Service brake pressure	3a	Was pedal pressure within specified limits?	____	___
	3b	Did brake system withstand applied pressure without failure?	____	___
Topping, sluing, and Hook speed	4	Does load line hook speed, boom topping speed and sluing speed comply with requirements?	____	___
Crane speed	5	Do average forward and rearward speeds comply with requirements?	____	___
Acceleration	6a	Does average acceleration comply with the specified requirements?	____	___
Acceleration response	6b	Is acceleration response within specified limits?	____	___
Slope ascension	7a	Does crane comply with the slope ascension requirements?	____	___
Parking brake	7b	Does parking brake meet requirements?	____	___
Stopping distance	8	Does stopping distance meet requirements?	____	___
Stability	9	Does crane comply with stability requirements?	____	___
Lifting attachments	10	Do lifting attachments show any permanent deformation?	____	___
Wear of steering mechanism	11	Was free play of steering wheel before and after endurance test within specified performance test?	____	___
Structural load	12	Do crane components show any indication of deformity?	____	___

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Turning radius	13a	Does crane comply with turning radius requirements?.....	_____	___
Travel capacity	13b	Does crane carry rated load?.....	_____	___
Power train test	14	Does Power train comply with test requirements?.....	_____	___
Controls	15	Do all controls comply with requirements?.....	_____	___
Hook and cable	16	Do hook and cable comply with requirements?.....	_____	___
General	Any evidence of failure, excessive wear or deformation in any component of the crane, or any defects which might cause failure during normal operation?		_____	___

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POST-TEST INSPECTION FORM

CRANE, HYDRAULIC, MOBILE, DIESEL, RUBBER TIRES

Instructions: Examine each item below in sufficient detail to determine any defects or excessive wear which have caused failure in test or might cause failure in normal operation. If item is non-defective and requires no additional correction, place a check mark in column No. 1. If corrections are required, check column Nos. 2, 3, or 4 as applicable. Place remarks opposite the proper number of this form. If more space is required, attach separate sheet. Remarks should include details of adjustments, repairs, or replacements indicated by check marks above.

Description	(1) No Corrections Required	(2) Adjust	(3) Repair	(4) Replace
1. Engine				
a. Ignition system				
b. Alternator				
c. Regulator				
d. Starting system				
e. Fuel system				
f. Cooling system				
g. Exhaust system				
2. Transmission and Drive Assembly				
a. Differential				
b. Drive axles				
c. Drive shaft and universal				
d. Lubrication fittings				
e. Torque converter				
3. Hydraulic System				
a. Hydraulic pumps				
b. Pump (storage) tank				
c. Lines and fittings				
d. Valves				
e. Controls and levers				
4. Structure (deformation)				
a. Chassis and frame				
b. Boom				
c. Boom supporting structure				
d. Hoisting mechanism				
e. Steer Axles				
f. Wheel assembly				
g. Springs				
5. Steering Assembly				
6. Parking Brake				
7. Service Brake				
8. Controls/Instrumentation				
9. Batteries				
10. Accessories				
11. Workmanship				
12. Treatment and Painting				
13. Repair/Ease of Maintenance				

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POST-TEST INSPECTION FORM (Cont'd)

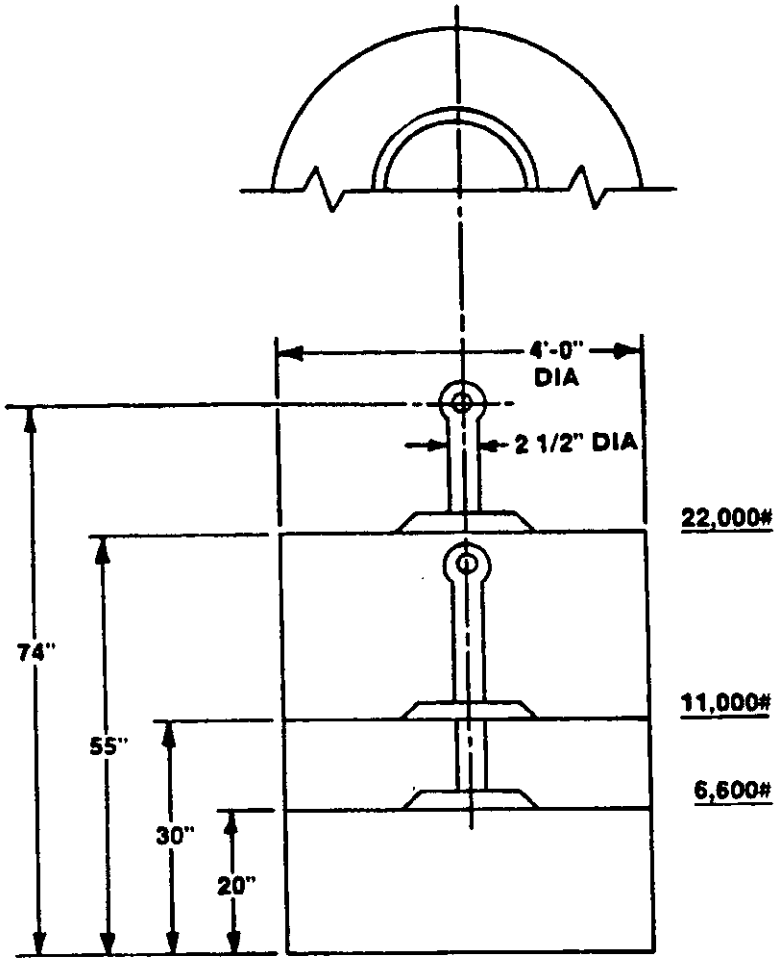
Description	(1) No Corrections Required	(2) Adjust	(3) Repair	(4) Replace
14. Technical Manual (Completeness),				
15. Cable				
16. Cable clamps				
17. Hook and safety catch				
18. Limit stops				

Crane Mfr. _____ Date _____
Model _____ Test Location _____
Serial No. _____ Mechanic _____
Maximum lift height _____ Payroll No. _____

Custodian:
Army - GL
Navy - SA
Air Force - 99

Preparing Activity
NAVY SA

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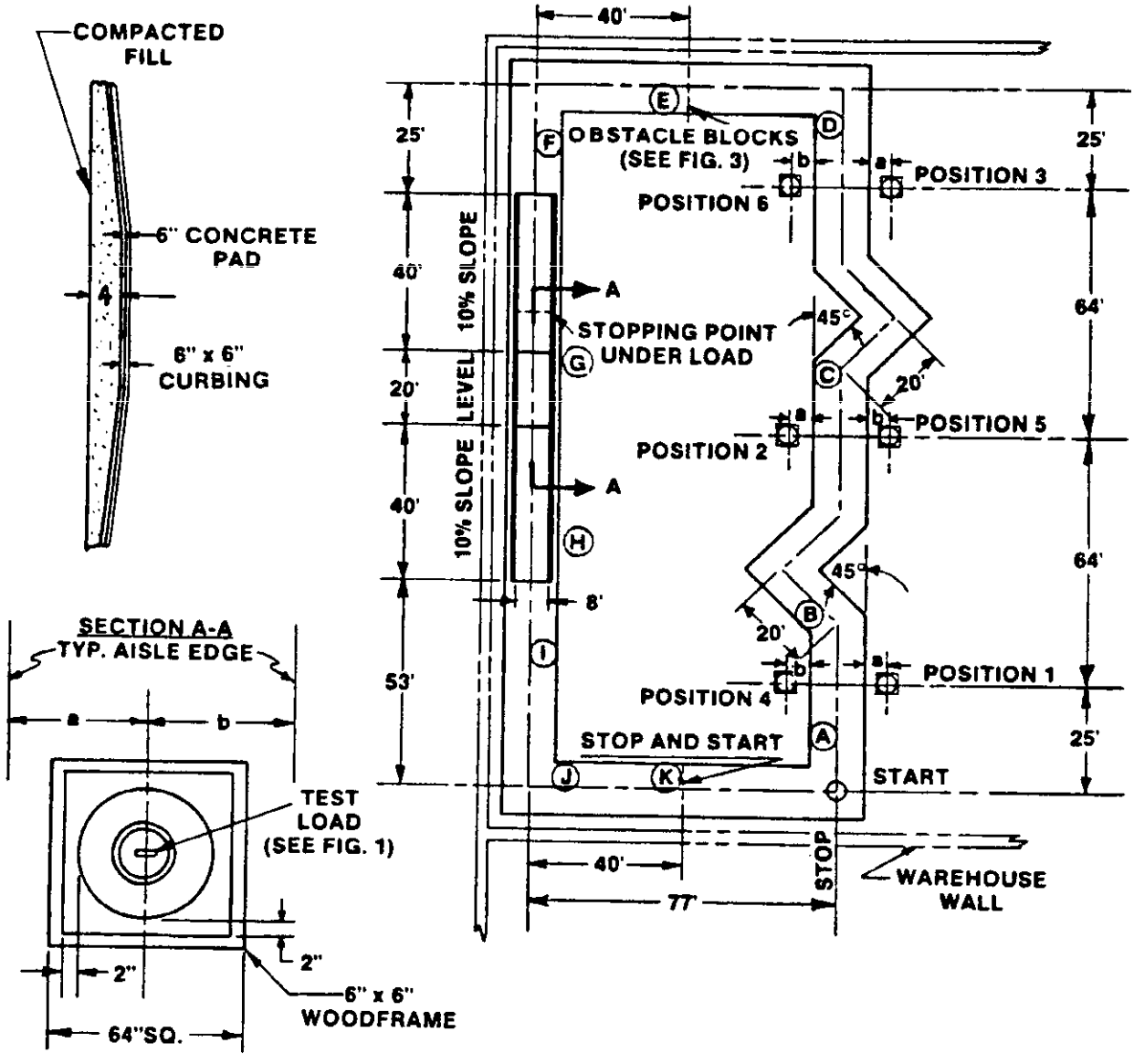


NOTES:

1. LOADS ARE 10%IN EXCESS OF THE APPLICABLE RATED LOADS FOR RESERVE STABILITY TESTS.
2. THE LOAD CONFIGURATION IS NOT MANDATORY, HOWEVER THE APPLICABLE MAGNITUDE, STATED ABOVE, IS REQUIRED FOR THE STABILITY TEST ONLY. FOR TEST COURSE PURPOSES THE CRANE TRUCK RATED LOADS ARE TO BE USED.
3. THE LOAD SPACE DIMENSIONS STATED, SHALL NOT BE GREATER THAN AS SHOWN ABOVE.

FIGURE 1 — CRANE STABILITY TEST LOADS

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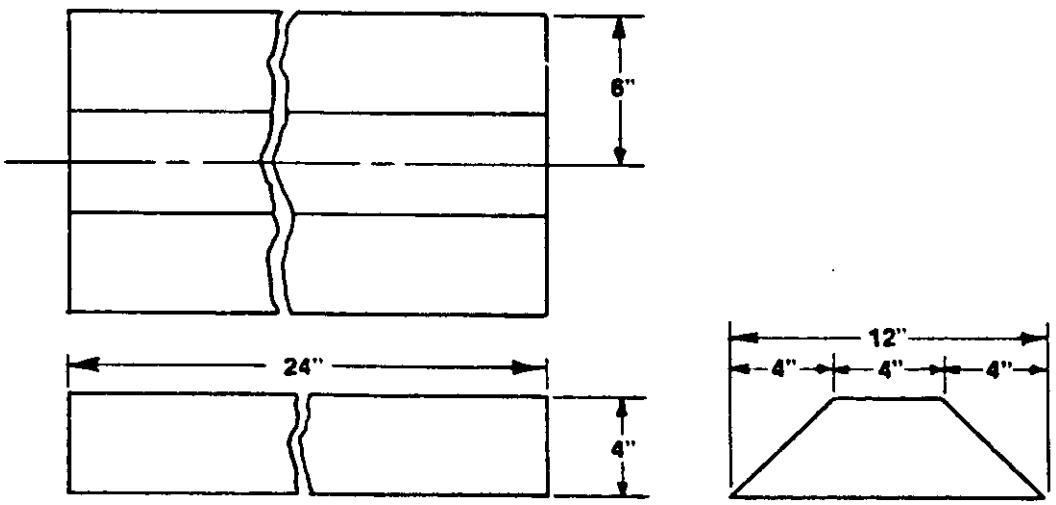
NOTES:

1. a — APPLICABLE OUTER DISTANCE SPECIFIED UNDER PARAGRAPH 1.2 OF GOVERNING SPECIFICATION WHEN CRANE IS ADJACENT TO LOAD POSITION.
   b — APPLICABLE INNER DISTANCE SPECIFIED UNDER PARAGRAPH 1.2 OF GOVERNING SPECIFICATION WHEN CRANE IS ADJACENT TO LOAD POSITION.
2. A TEST SECTION DESIGNATION.
3. MINIMUN LIGHTING AT OPERATING LEVEL TO BE 5 FOOT CANDLES.
4. TEST COURSE & RAMP TO BE FINE - FLOAT FINISH CONCRETE.
5. TEST COURSE SHOWN IN A STANDARD 200' x 600' NAVY WAREHOUSE.

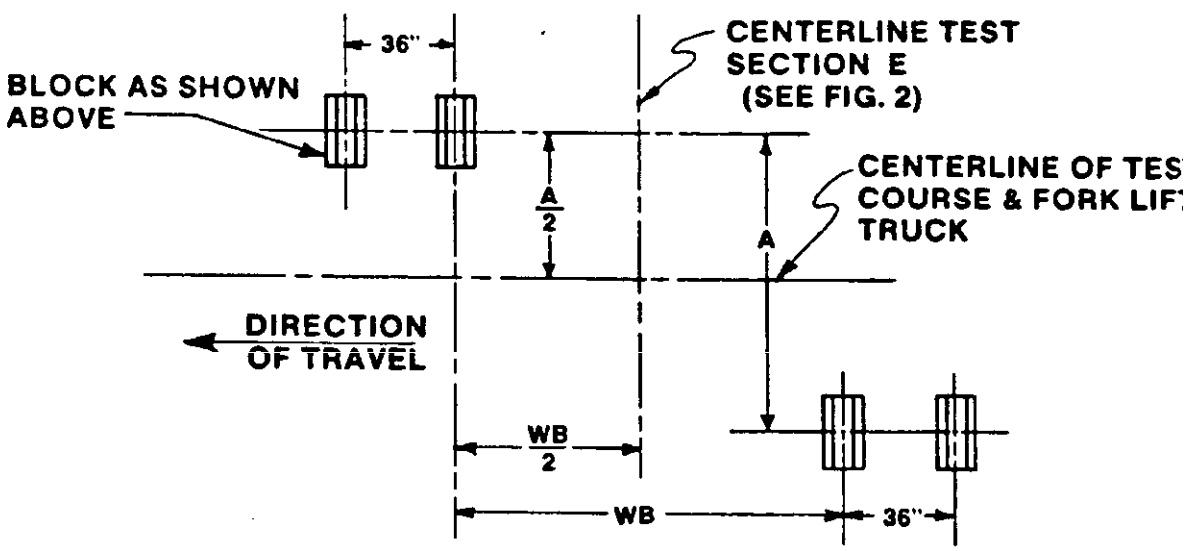
FIGURE 2 — INDOOR TEST COURSE FOR CRANE TRUCKS

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OBSTACLE BLOCK CONSTRUCTION



LAYOUT FOR OBSTACLE COURSE



NOTES:

1. "A" TREAD OF DRIVING WHEELS
2. "WB" WHEEL BASE OF TRUCK UNDER TEST

FIGURE 3 — DETAIL OF OBSTACLE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-261B	Hydraulic, Mobile, Diesel, Rubber Tires

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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